The University Of Azad Jammu & Kashmir, Muzaffarabad



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Course: **Software Construction**

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Lab#3

Q1: ATM cash withdrawal

Introduction:

A sequence diagram shows how objects interact with each other over time. This diagram shows how data is processed when a customer withdraws cash from an ATM.

Objects:

Customer, ATM, Bank Server, Account DB, Cash Dispenser.

Explanation:

- The customer inserts the card. The ATM asks for the PIN and the customer enters it.
- The ATM sends the card number and PIN to the bank server, which checks them against the account database.
- If the PIN is valid, the ATM asks for the withdrawal amount.
- The bank server checks the balance. If the balance is greater than or equal to the amount (alt fragment), the account is debited, the ATM dispenses cash, prints the receipt and ejects the card.
- If the balance is less than the amount, the transaction is declined, a message is shown and the card is ejected.

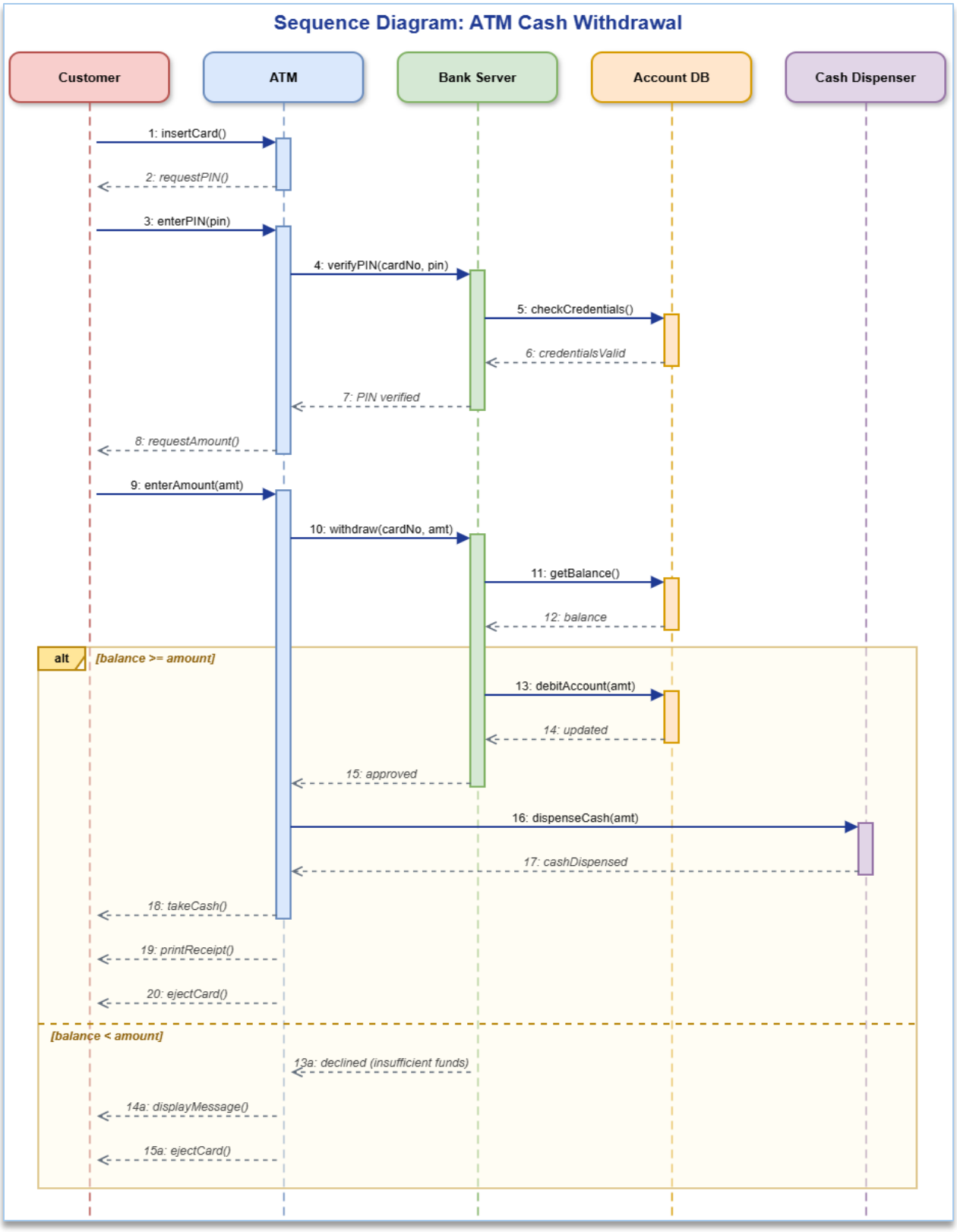


Fig: Sequence Diagram of ATM Cash Withdrawal

Notation:

- **Solid arrow** = request/call
- **Dashed arrow** = return/response
- **Vertical dashed line** = lifeline
- **alt box** = either/or (if/else) situation

Q2: Online Ticket Reservation System

Introduction:

A sequence diagram shows how objects interact with each other in time order. This diagram shows how a customer searches for a trip, reserves a seat, pays online and receives a ticket through an online ticket reservation system.

Objects (Lifelines):

- Customer: the person who books the ticket.
- Web Interface: the website or app the customer uses.
- Reservation System: the main system that handles the booking logic.
- Database: stores trips, seats and bookings.

- Payment Gateway: processes the online payment.

- Email Service: sends the ticket to the customer.

Explanation:

1. Searching trips: The customer enters the origin, destination and date on the web interface. The interface passes the request to the reservation system, which queries the database. The database returns the trip list and the web interface displays the available trips to the customer.

2. Selecting a seat: The customer selects a trip and a seat. The reservation system locks that seat in the database so that nobody else can book it at the same time, and the database confirms the lock.

3. Entering details and payment: The web interface asks for passenger details. The customer submits the details and payment information, and the reservation system sends the payment amount to the payment gateway. The gateway processes it and returns the payment result.

4. Alternative flow (alt fragment): The next steps depend on the payment result.
    - If payment is successful: The reservation system confirms the booking in the database, and the database saves it. The email service then sends the ticket to the customer. The web interface shows the booking confirmation and the ticket.

    - If payment fails: The reservation system releases the locked seat in the database, so it becomes available again. The web interface shows a payment failed error message to the customer.

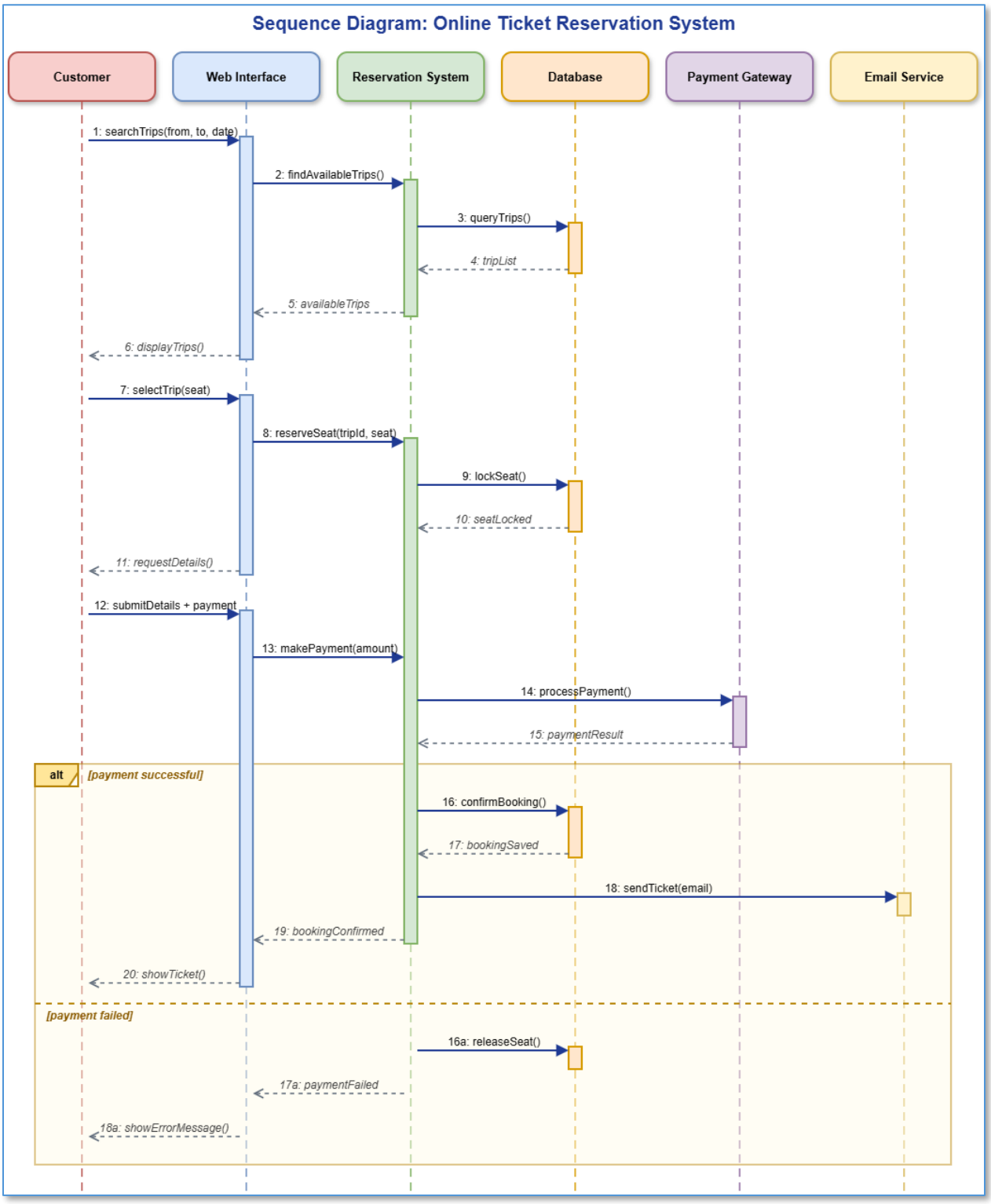


Fig: Online Ticket Reservation System

Notation used:

- **Solid arrow:** a request (call) from one object to another.
- **Dashed arrow:** a response (return) to a call.
- **Vertical dashed line:** the lifeline of an object.
- **Thin vertical bar:** an activation, when the object is busy.
- **alt box:** alternative flow, where only one of the two paths happens.

Conclusion:

This diagram shows the complete booking process from searching to ticket delivery, including the case where the payment fails.

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